Experiment: 08 **User Session Tracking System using Servlet**

Aim:

To maintain the transactional history of any user using various session tracking mechanism (Cookies, HTTP Session)**.**

Description:

Servlet Session:

- Keeping track of a user's information while they move between different pages of your web application.

Two Main Techniques

1. **Cookies**
   - Small pieces of data stored in the user's browser.
   - Example: You set a cookie with the username.
   - When the user visits another page, the browser automatically sends the cookie back to the server.
2. **HttpSession**
   - Managed by the server (Tomcat).
   - When a user first visits, Tomcat creates a session object and gives the browser a session ID (usually stored in a cookie).
   - You can store attributes (like username, cart items) in the session object.

Project Structure:

Create all the folders in the location

C:/Program Files/Apache Software Foundation/Tomcat 10.1/webapps/

```
shopping-cart-app-session/
├── src/
│   └── SessionServlet.java          # Java source files
├── lib/                             # Extra libraries
│   ├── mysql-connector-j-9.6.0.jar # (Mostly in Desktop)
│   └── servlet-api.jar        # (C:\Program Files\Apache Software Foundation\Tomcat 10.1\lib)
├── index.html                       # Frontend form
└── WEB-INF/
    ├── web.xml                      # Servlet config
    └── classes/                     # Compiled .class files go here
```

Source Code

1. Create **web.xml** in the folder shopping-cart-app-session/ WEB-INF/**:**

```
<servlet>
    <servlet-name>SessionServlet</servlet-name>
    <servlet-class>SessionServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>SessionServlet</servlet-name>
    <url-pattern>/session</url-pattern>
  </servlet-mapping>
</web-app> >
```

2. Create the Servlet Session **(SessionServlet.java)** in the folder shopping-cart-app-session/src**:**

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class SessionServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");

        // Cookie example
        Cookie userCookie = new Cookie("username", username);
        userCookie.setMaxAge(60); // 1 minute
        response.addCookie(userCookie);

        // HttpSession example
        HttpSession session = request.getSession();
        session.setAttribute("username", username);

        out.println("<h3>Welcome, " + username + "</h3>");
        out.println("<p>Cookie and HttpSession have been set.</p>");
        out.println("<a href='session'>Go to Session Page</a>");
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        // Read cookie
        String cookieUser = null;
        Cookie[] cookies = request.getCookies();
        if (cookies != null) {
            for (Cookie c : cookies) {
                if ("username".equals(c.getName())) {
                    cookieUser = c.getValue();
                }
            }
        }
```

// Read session

```
        HttpSession session = request.getSession(false);
        String sessionUser = (session != null) ? (String)
session.getAttribute("username") : null;

        out.println("<h3>Session Page</h3>");
        out.println("<p>Cookie user: " + cookieUser + "</p>");
        out.println("<p>Session user: " + sessionUser + "</p>");
    }
}
```

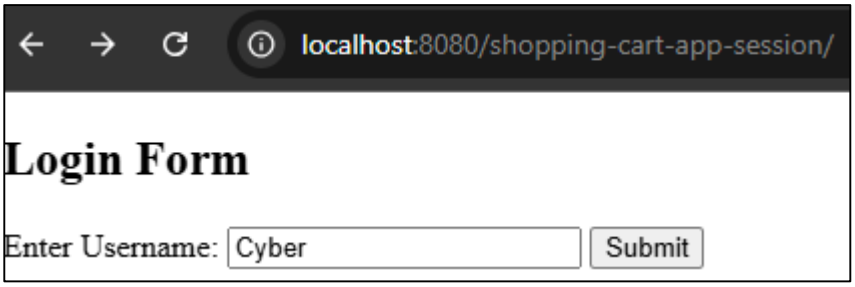
3. Create a Frontend Form **(index.html)** in the folder shopping-cart-app-session **:**

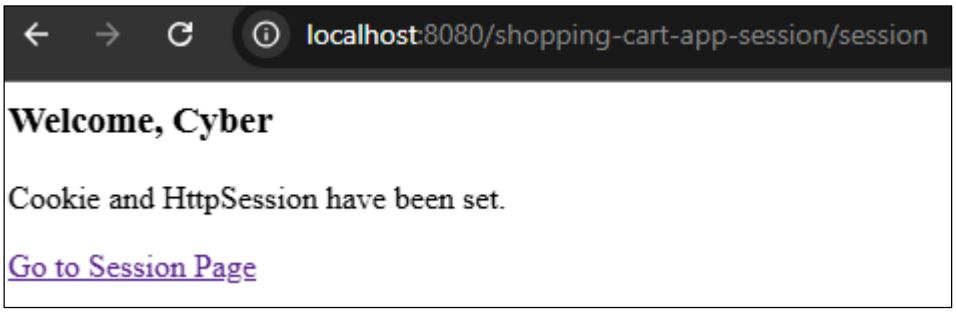
```
<html>
<head>
   <title>Session Management</title>
</head>
<body>
   <h2>Login Form</h2>
   <form action="session" method="post">
      <label for="username">Enter Username:</label>
      <input type="text" id="username" name="username" required>
      <input type="submit" value="Submit">
   </form>
</body>
</html>
```

How to Execute:

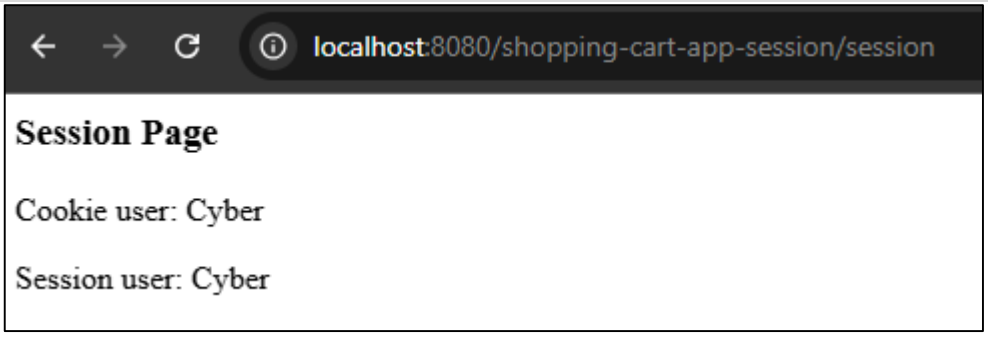
1. Save all your work in the location as per project structure C:/Program Files/Apache Software Foundation/Tomcat 10.1/webapps/**shopping-cart-app-session/**
2. Compile Your Servlet in VS Code in the terminal (Press **Ctrl + `)**
   >javac -cp "C:/Program Files/Apache Software Foundation/Tomcat 10.1/lib/*" -d "C:/Program Files/Apache Software Foundation/Tomcat 10.1/webapps/**shopping-cart-app-session**/WEB-INF/classes" "C:/Program Files/Apache Software Foundation/Tomcat 10.1/webapps/**shopping-cart-app-session/src/SessionServlet.java"**
3. After compilation, check the folder: shopping-cart-app/WEB-INF/classes/**SessionServlet**.class
4. **Start Tomcat**
   Press Windows logo -> type services -> choose Apache Tomcat 10.1-> Right click -> start
5. **Run in Browser**
   http://localhost:8080/shopping-cart-app-session

Output:





When you click Go to Session Page



Viva Questions:

1. What is the purpose of session management in servlets?
2. How does HttpSession differ from cookies in storing user data?
3. How do you set and retrieve a cookie in a servlet?
4. Why do we use session.setAttribute() and session.getAttribute() methods?
5. How does the server identify which session belongs to which user?